RaFTS: Open-source analytics tool for hydrologic model regionalization & decision making

OWP | OFFICE OF WATER PREDICTION

Enabling community contributions in formulation and hydrologic signature regionalization for continental hydrologic modeling

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NOAA

Challenge:

- NextGen v4: >800k CONUS catchments + OCONUS
- Which hydrologic models, where?

RaFTS

- **R**egionalization **a**nd **F**ormulation **T**esting & **S**election
- Predicts response variables based on catchment-wide descriptors
- Response variables:
    - Model performance (e.g. KGE)
    - Hydrologic signatures
    - Process sensitivities

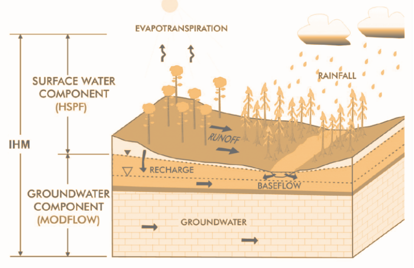
Sections

1. How RaFTS works
    a. Methods
    b. Application to predicting NNSE
2. Process Sensitivity
    a. Background
    b. Examples & application
3. RaFTS & community

RaFTS Training

TRAINING INPUTS

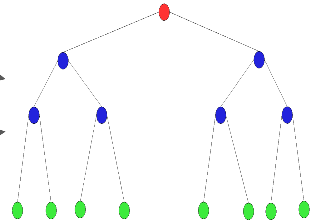
Model evaluation metrics



- *NNSE*
- *KGE*
- etc.

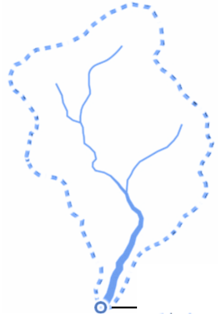
Response

TRAIN ALGO



e.g. Random Forest Regression

Catchment/climate attributes



- *precip patterns*
- *land cover*
- *soils*
- *topography*
- *etc.*

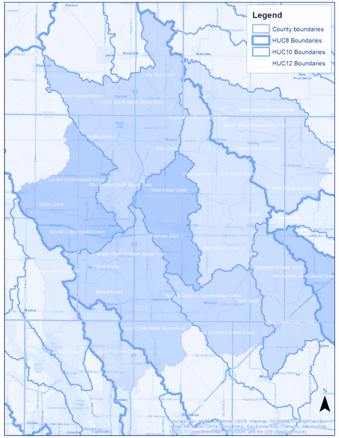
Predictors

RaFTS Prediction

PREDICTION INPUT

Predictors

Catchment attributes



- *precip patterns*
- *land cover*
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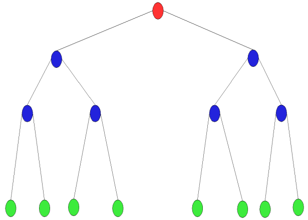
- *etc.*

TRAINED ALGO



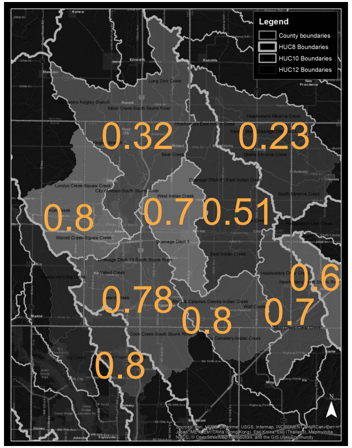
Random Forest Regression

PREDICTION OUTPUT

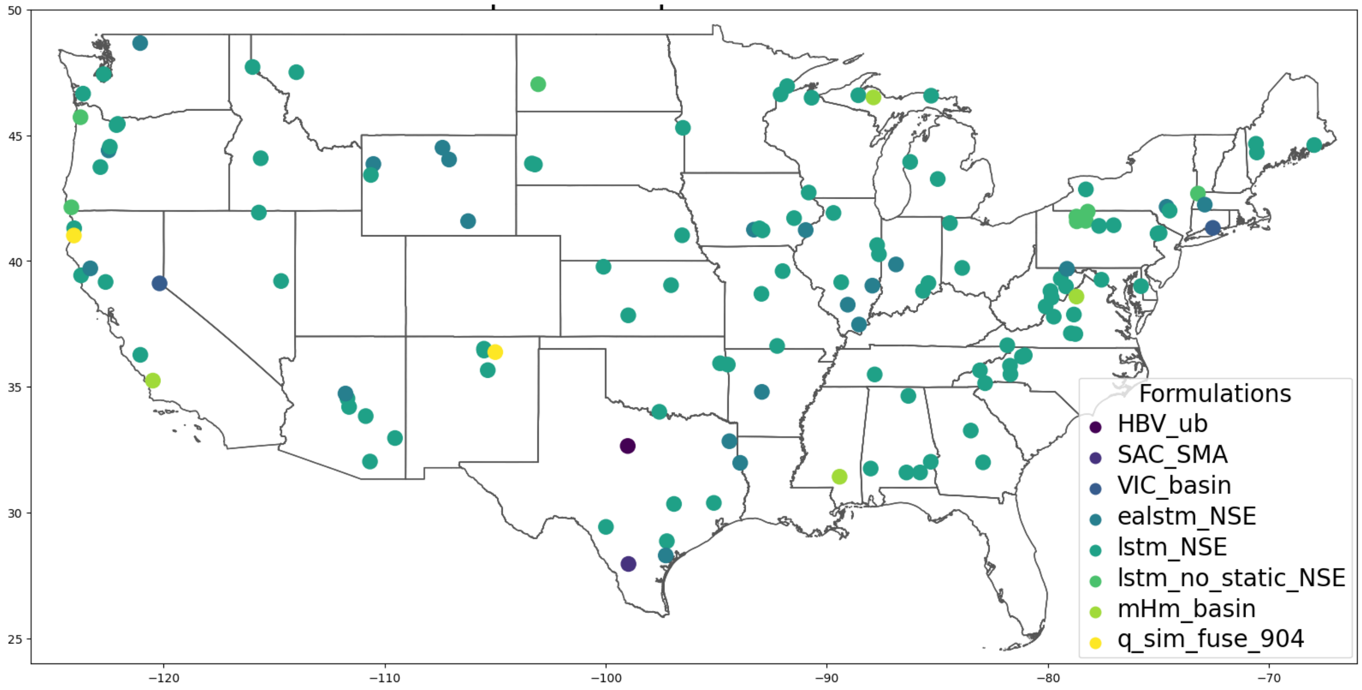
Response

Predicted model evaluation metric

NNSE:



RaFTS Application: NNSE



Formulations with best-predicted NNSE

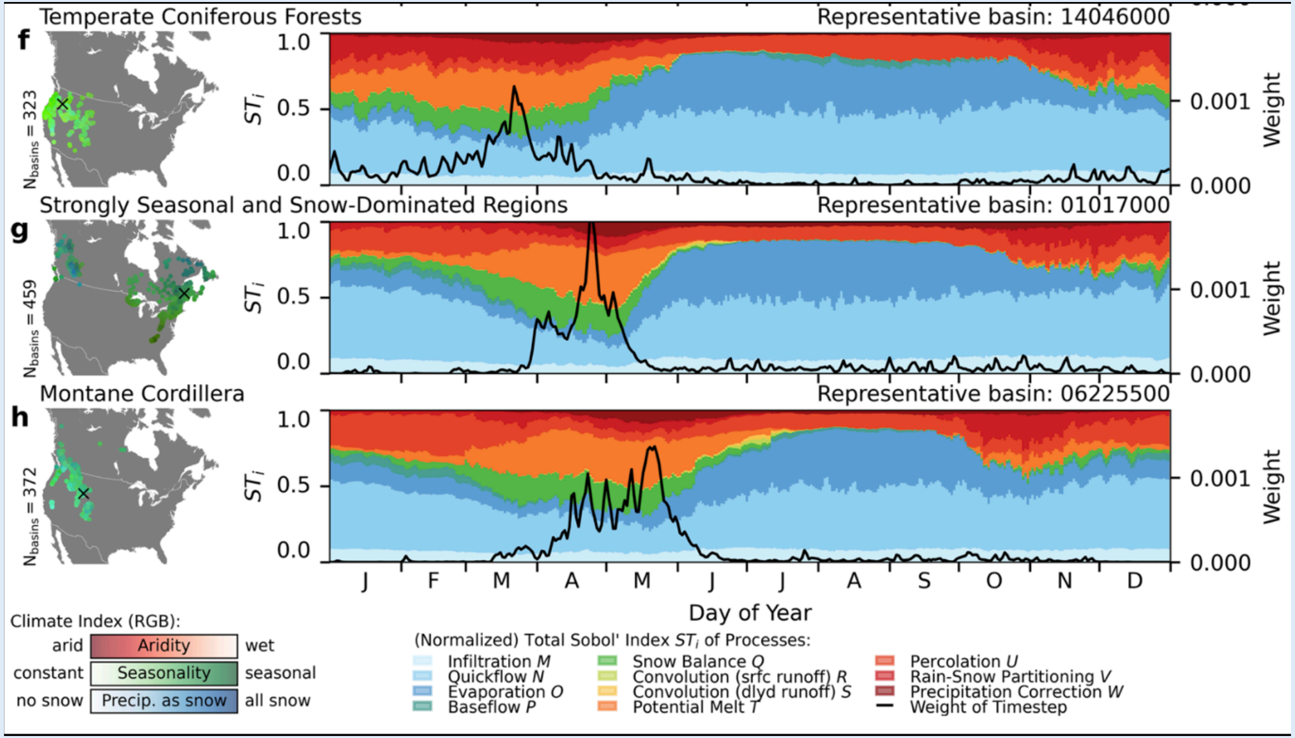
Which hydrologic model is predicted to perform the best at each location?

Section 2

Process Sensitivity

A process-based strategy for formulation selection & calibration

Process Sensitivity to Streamflow

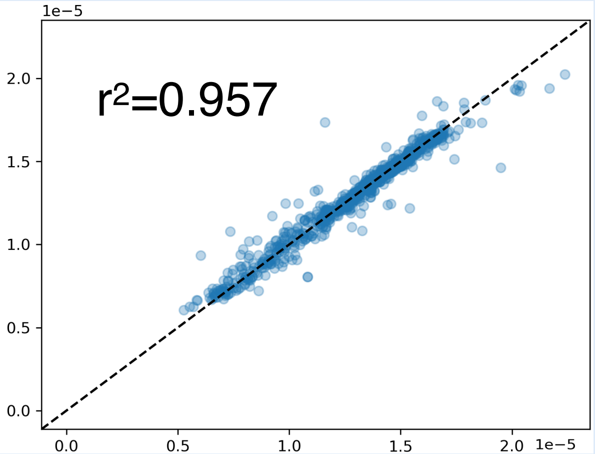


- Quantifies how ***sensitivity* of hydrologic model predictions of streamflow** to different **hydrologic processes**
- Uses 3000 basins across N. America

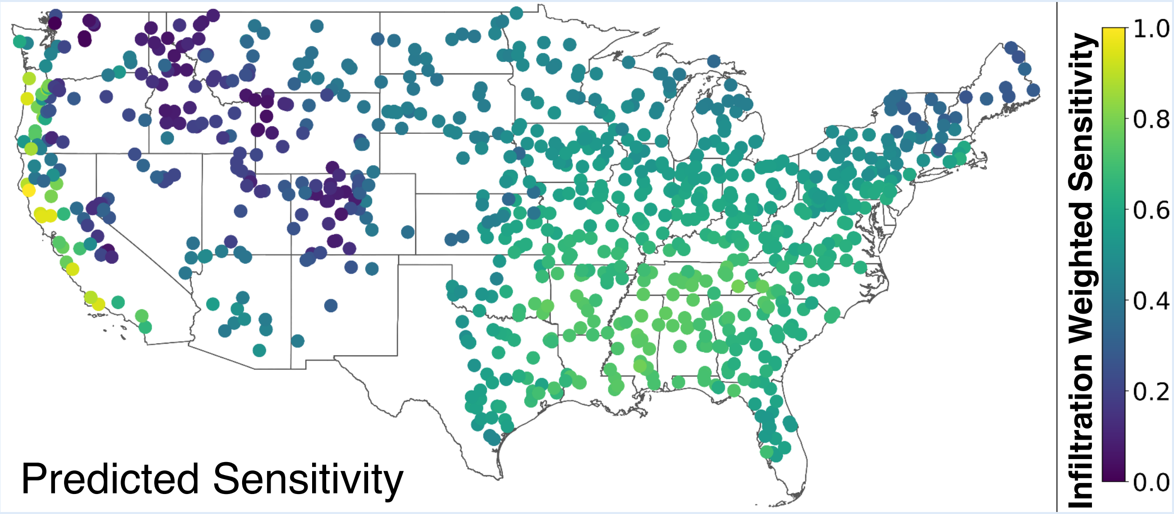
Mai, J., Craig, J.R., Tolson, B.A., Arsenault, R., 2022. The sensitivity of simulated streamflow to individual hydrologic processes across North America. Nature Communications. https://doi.org/10.1038/s41467-022-28010-7; Figure 4f-h

RaFTS Predicted *Infiltration* Process Sensitivity

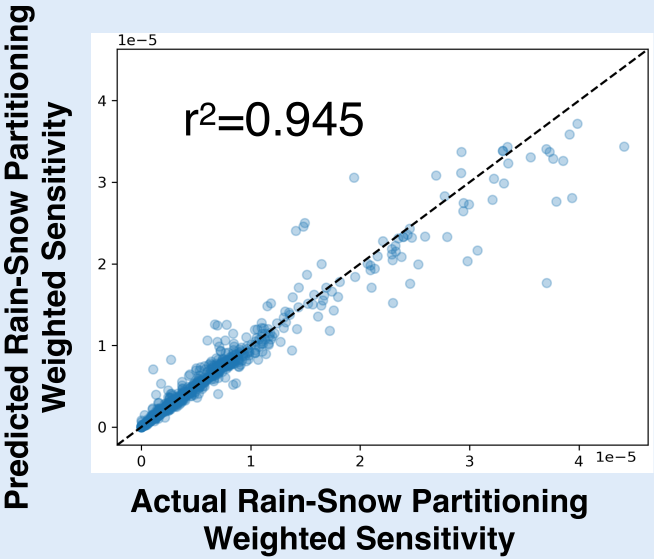
Predicted Infiltration Weighted Sensitivity

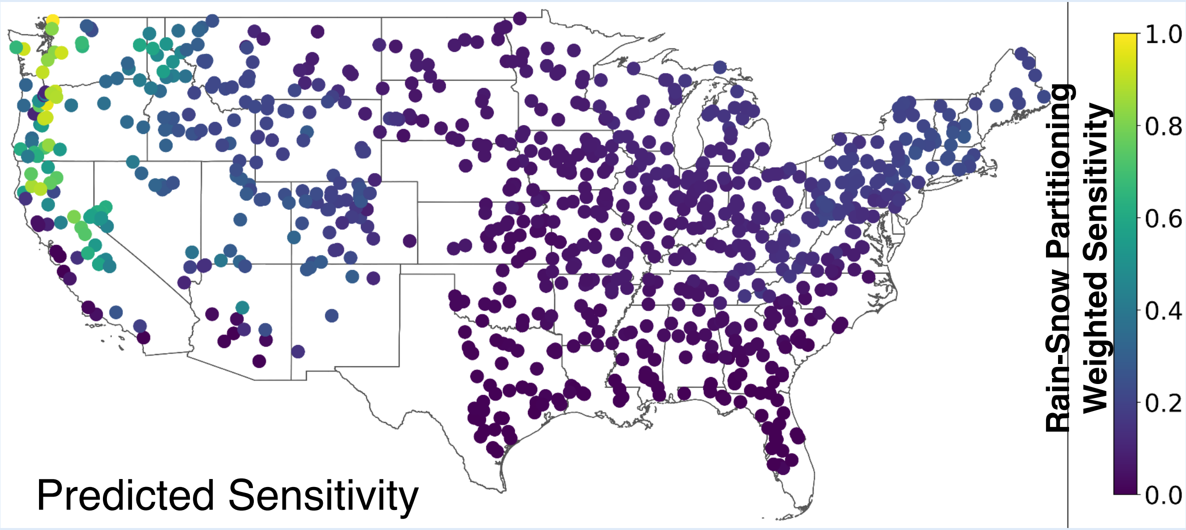


Actual Infiltration Weighted Sensitivity



RaFTS Predicted *Rain-Snow Partitioning* Sensitivity



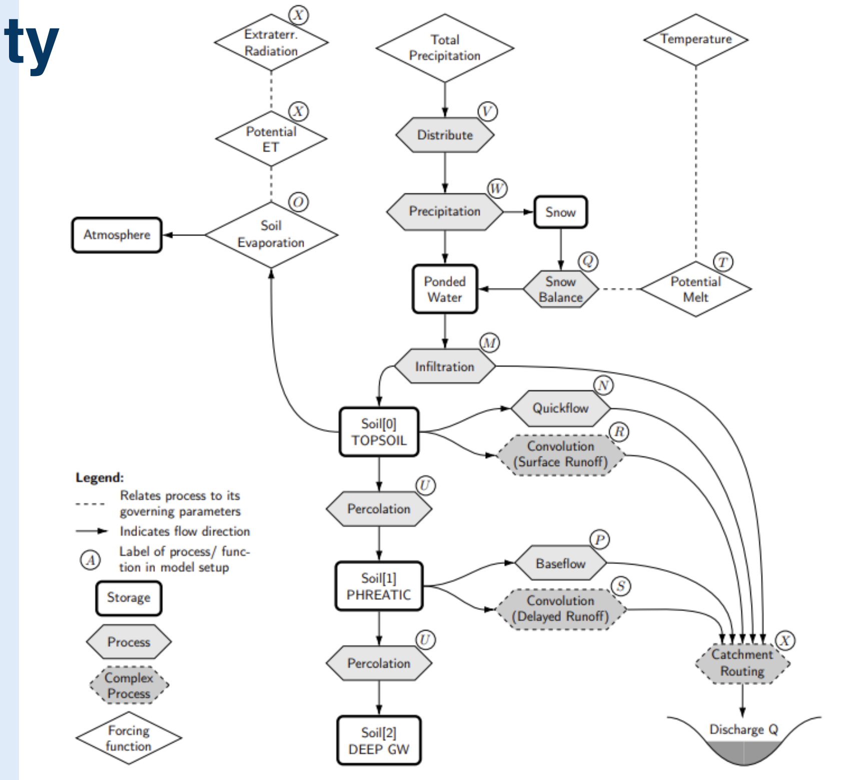


Application of Process Sensitivity

Sobol' Sensitivity: A measure of which processes are important

- Which hydrologic formulations represent the important processes?
  ↳ Decision support for formulation selection

- Which parameters correspond to the important processes?
  ↳ Simplify calibration



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Mapping Sensitivities to NextGen Modules

Sec 2b

Process	Noah-OM	CFE (X and S)	TOPMODEL
Infiltration	0	19	2
Quickflow	0	13	11
Evaporation	4	0	0
Snow Balance	3	0	0
Convolution (srfc)	0	9	1
Potential Melt	2	0	0
Percolation	0	12	3
Rain-Snow Partitioning	3	0	0
Precipitation Correction	0	0	0
Baseflow	0	6	1
Convolution (subsurface)	0	3	1

≥ 1 parameter(s)   0: no model parameters

RaFTS & community resources

- Community-shared data improves decision making!

  - To date, RaFTS uses publicly available data

- Open source RaFTS development for community use:

  https://github.com/NOAA-OWP/formulation-selector

Acknowledgments

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https://github.com/NOAA-OWP/formulation-selector

Thank You!

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